ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Electricity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** coulomb	1 C is the charge of about 6×10^{18} electrons.
2	**(a)** directly proportional to the potential difference, at constant temperature	V = IR.
3	**(b)** R1 + R2 + R3	Rs = R1 + R2 + R3.
4	**(b)** ampere	1 A = 1 C/s.
5	**(d)** ohm	1 ohm = 1 V/A.
6	**(b)** current	Same current.
7	**(d)** ammeter connected in series	An ammeter has a very low resistance.
8	**(b)** a straight line through the origin	Slope = R.
9	**(c)** potential difference	Same voltage.
10	**(d)** voltmeter connected in parallel	A voltmeter has a very high resistance.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Current when 1 coulomb of charge flows through a point in 1 second (2).	2
12	Length and area of cross-section (1); Nature of the material and temperature (1).	1+1
13	P = VI (1); P = I^2 R = V^2 / R (1).	1+1
14	Work done to move a unit charge from one point to another (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Q = I t = 0.5 x 600 = 300 C (1); n = Q / e (1); n = 300 / 1.6×10^{-19} = 1.875×10^{21} (1).	1+1+1
16	Rs = 2 + 3 + 5 = 10 ohm (1); I = V / R (1); I = 0.6 A (1).	1+1+1
17	Filament has very high resistance, so heat produced is large (1); Heat H = I^2 R t (1); Connecting wires have low resistance (1).	1+1+1
	OR: W = V Q (1); W = 12 x 2 (1); W = 24 J (1).	1+1+1
18	1/R = 1/20 + 1/30 + 1/60 (1); 1/R = 6/60 = 1/10 (1); R = 10 ohm (1).	1+1+1
	OR: Energy = 2 x 3 = 6 kWh (1); 1 unit = 1 kWh (1); Cost = 6 x 5 = Rs 30 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	V is proportional to I at constant temperature (1); Connect cell, ammeter, resistor and key in series and a voltmeter across the resistor (1); Take readings for different numbers of cells (1); The V-I graph is a straight line through the origin (1); The slope of the graph gives the resistance (1).	1+1+1+1+1
	OR: Work done to move charge Q across V is W = V Q (1); Q = I t (1); W = V I t (1); Using V = I R, W = I^2 R t (1); This energy is dissipated as heat H (1).	1+1+1+1+1
20	Same current I flows through each (1); V = V1 + V2 + V3 = I (R1 + R2 + R3) (1); Rs = R1 + R2 + R3 (1); Rs = 3 + 4 + 5 (1); Rs = 12 ohm (1).	1+1+1+1+1
	OR: Electric heater, toaster, iron (1); Electric bulb (1); Fuse works by heating its wire (1); Fuse wire has low melting point (1); It melts when current exceeds the limit and breaks the circuit (1).	1+1+1+1+1

Mistakes to avoid:

- Adding the reciprocals incorrectly: for resistors in parallel $\frac{1}{R_p} = \sum \frac{1}{R}$ and the final answer is R_p, not the sum of reciprocals.
- Connecting an ammeter in parallel or a voltmeter in series; the ammeter goes in series and the voltmeter in parallel.
- Thinking that resistance is directly proportional to cross-sectional area; $R \propto \frac{1}{A}$ and $R \propto l$.
- Using $H = I^2Rt$ with t in minutes or hours; time must be in seconds (and energy in kWh needs power in kW and time in hours).

- Writing that in a series circuit the voltage across each resistor is the same; the **current** is the same in series and the **voltage** is the same in parallel.
- Forgetting that 1 kWh equals 3.6×10^6 J; and confusing the unit of resistivity (Ω m) with that of resistance (Ω).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Electricity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** directly proportional to its length	R is proportional to L.
2	**(b)** 1 ohm	1/Rp = 1/2 + 1/2.
3	**(b)** 1 joule of work is done to move 1 coulomb of charge	V = W/Q.
4	**(c)** area of cross-section	R = rho L / A.
5	**(a)** 9 ohm	Add.
6	**(c)** Q = I t	I = Q/t.
7	**(b)** ohm metre	rho = R A / L.
8	**(b)** less than the smallest resistance	More paths reduce R.
9	**(a)** Copper	Free electrons are present in metals.
10	**(d)** higher than that of its constituent metals	Alloys have higher resistivity.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Resistivity of alloys is higher (1); They do not oxidise readily at high temperature (1).	1+1
12	H = I^2 R t (1); Heat depends on the square of current, resistance and time (1).	1+1
13	Potential difference across a conductor is proportional to the current (1); Temperature and other conditions remain constant (1).	1+1
14	Copper has low resistivity (1); It is a good conductor and ductile (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	V = I R (1); V = 0.5 x 20 (1); V = 10 V (1).	1+1+1
16	Series: same current; parallel: same potential difference (1); Series: R increases; parallel: R decreases (1); Series: one break stops all; parallel: others still work (1).	1+1+1
17	High melting point (1); High resistivity (1); It does not oxidise readily in an inert gas (1).	1+1+1
	OR: Cell, key, resistor and ammeter are in series (1); Voltmeter is connected in parallel with the resistor (1); Readings of V and I are noted and V/I is found to be constant (1).	1+1+1
18	Parallel: 6 x 3 / 9 = 2 ohm (1); Total = 2 + 4 (1); Total = 6 ohm (1).	1+1+1
	OR: Current, resistance and time (1); H = I^2 R t (1); Also equals V I t (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Same potential difference V across each (1); Total current I = I1 + I2 (1); 1/Rp = 1/R1 + 1/R2 (1); 1/Rp = 1/12 + 1/4 = 4/12 (1); Rp = 3 ohm (1).	1+1+1+1+1
	OR: 1/R = 1/2 + 1/3 + 1/6 = 1 (1); R = 1 ohm (1); I = 6 / 1 = 6 A (1); P = V I = 36 W (1); Individual currents 3 A, 2 A, 1 A (1).	1+1+1+1+1
20	1/R = 1/40 + 1/60 = 5/120 (1); R = 24 ohm (1); I1 = 12 / 40 = 0.3 A (1); I2 = 12 / 60 = 0.2 A (1); Total current 0.5 A (1).	1+1+1+1+1
	OR: R = rho L / A (1); Increases with length and decreases with area (1); Depends on the material and temperature (1); Nichrome has much higher resistivity than copper (1); So the nichrome wire has a greater resistance (1).	1+1+1+1+1

Mistakes to avoid:

- Adding the reciprocals incorrectly: for resistors in parallel $\frac{1}{R_p} = \sum \frac{1}{R}$ and the final answer is R_p, not the sum of reciprocals.
- Connecting an ammeter in parallel or a voltmeter in series; the ammeter goes in series and the voltmeter in parallel.
- Thinking that resistance is directly proportional to cross-sectional area; $R \propto \frac{1}{A}$ and $R \propto l$.
- Using $H = I^2Rt$ with t in minutes or hours; time must be in seconds (and energy in kWh needs power in kW and time in hours).

- Writing that in a series circuit the voltage across each resistor is the same; the **current** is the same in series and the **voltage** is the same in parallel.
- Forgetting that 1 kWh equals 3.6×10^6 J; and confusing the unit of resistivity (Ω m) with that of resistance (Ω).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Electricity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** Each gets the same voltage and works independently	Parallel connection.
2	**(d)** the positive terminal to the negative terminal outside the cell	Opposite to the direction of electron flow.
3	**(a)** Nichrome	High resistivity and does not oxidise easily.
4	**(c)** I^2 R t	Joule's law.
5	**(b)** 1.6×10^{-19} C	Magnitude of the charge on an electron.
6	**(b)** Silver	Silver is the best conductor.
7	**(b)** watt	1 W = 1 J/s.
8	**(c)** a cell or battery	A source of emf.
9	**(c)** 4R	R = rho (2L)/(A/2).
10	**(d)** 3.6×10^6 J	1 kW x 3600 s.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It melts when the current exceeds the safe limit (1); It protects the appliance (1).	1+1
12	Ammeter has low resistance, so the current is not changed (1); Voltmeter has high resistance, so it draws almost no current (1).	1+1
13	Resistance is proportional to length (1); It becomes double (1).	1+1
14	Kilowatt hour (1); 3.6×10^6 J (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Longer path means more collisions of electrons (1); Larger area gives more paths for electrons (1); R = rho L / A (1).	1+1+1
16	I = P / V = 1100 / 220 = 5 A (1); R = V / I (1); R = 44 ohm (1).	1+1+1
17	Same current flows through all appliances (1); If one fails the entire circuit breaks (1); Voltage is divided among appliances (1).	1+1+1
	OR: rho = R A / L (1); rho = 4 x 0.5×10^{-6} / 2 (1); rho = 1×10^{-6} ohm m (1).	1+1+1
18	H = I^2 R t (1); H = 4 x 10 x 30 (1); H = 1200 J (1).	1+1+1
	OR: R = V^2 / P = 48400 / 100 (1); R = 484 ohm (1); I = P / V = 0.45 A (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Resistance of a conductor of unit length and unit area (1); Depends on material and temperature, not on size (1); rho = R A / L (1); rho = 1.5×10^{-6} / 1 (1); rho = 1.5×10^{-6} ohm m (1).	1+1+1+1+1
	OR: Energy per day = 2 x 2 = 4 kWh (1); In 30 days = 120 kWh (1); 120 units (1); Cost = 120 x 6 (1); Rs 720 (1).	1+1+1+1+1
20	I = 60/220 = 0.27 A (1); R = 220^2 / 60 = 806.7 ohm (1); Energy = 0.06 x 5 = 0.3 kWh (1); 0.3 x 3.6×10^6 (1); = 1.08×10^6 J (1).	1+1+1+1+1
	OR: Parallel part = 6 x 3 / 9 = 2 ohm (1); Total R = 4 + 2 = 6 ohm (1); I = 12 / 6 = 2 A (1); Voltage across 4 ohm = 8 V (1); Voltage across parallel part = 12 - 8 = 4 V (1).	1+1+1+1+1

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- Writing that in a series circuit the voltage across each resistor is the same; the **current** is the same in series and the **voltage** is the same in parallel.
- Forgetting that 1 kWh equals 3.6×10^6 J; and confusing the unit of resistivity (Ω m) with that of resistance (Ω).

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